

Cite this: *J. Mater. Chem. C*,  
2026, 14, 8715

## Compact and functional co-self-assembled monolayers for efficient inverted wide-bandgap perovskite solar cells

Xinying Ruan,<sup>a</sup> Jiahao Zhang,<sup>a</sup> Xiaolong Liu,<sup>\*b</sup> Wenli Shang,<sup>a</sup> Yuechen Li,<sup>id c</sup>  
Xiaojun Li,<sup>id c</sup> Ye Zou,<sup>c</sup> Ming-Hua Li,<sup>d</sup> Chunru Wang,<sup>id c</sup> and Fuwen Zhao<sup>id \*a</sup>

Wide-bandgap perovskite solar cells (PSCs) have garnered significant attention owing to their potential for use in tandem structures, indoor photovoltaics, and related applications. However, wide-bandgap PSCs still face notable challenges, including substantial open-circuit voltage loss and inferior perovskite film quality, mainly due to the imperfection of the hole transport layer at the buried interface. Herein, we synthesize a cyanation-functionalized self-assembled monolayer material, CN-4PACz, and blend it with Me-4PACz to construct a co-assembled monolayer material (co-SAM) as the hole transport layer in wide-bandgap PSCs. The cyano group attached onto CN-4PACz endows it with a deeper highest occupied molecular orbital energy level, larger molecular dipole moment and stronger adsorption energy on the perovskite. Thus, the co-SAM presents improved energy level alignment with the perovskite, higher surface energy, enhanced defect passivation capacity, increased electrical conductivity and more uniform morphology compared with Me-4PACz. The perovskite film deposited on co-SAM layers presents improved crystal quality with a larger grain size, more uniform morphology, and reduced interfacial defects, which help inhibit trap-assisted recombination and benefit charge transport and extraction in devices. Consequently, wide-bandgap PSCs having the co-SAM layer achieve an impressive power conversion efficiency of 18.41% with a higher open-circuit voltage of 1.339 V and a fill factor of 83.08%, outperforming the control device with Me-4PACz (17.45%). The efficiency of 18.41% is among the highest values for wide-bandgap (>1.86 eV) PSCs so far. This work highlights an efficient approach for developing high-performance self-assembled monolayer materials, offering a promising pathway toward advancing wide-bandgap PSCs and their application in tandem solar cells.

Received 12th March 2026,  
Accepted 19th March 2026

DOI: 10.1039/d6tc00781c

rsc.li/materials-c

## Introduction

Perovskite solar cells (PSCs) are regarded as some of the most promising candidates for next-generation photovoltaic techniques, owing to their splendid power conversion efficiency, tunable photoelectric properties and low manufacturing costs.<sup>1,2</sup> Among them, wide-bandgap PSCs have attracted considerable interest, due to their potential for application in tandem solar cells, indoor photovoltaics and related

technologies.<sup>3–6</sup> However, wide-bandgap PSCs generally suffer from large energy loss to generate relatively low open-circuit voltage ( $V_{OC}$ ), which is mainly ascribed to the high trap density in perovskite films, and the energy-level mismatch between the perovskite layer and the charge transport layer.<sup>7–9</sup> Moreover, it has been widely accepted that the quality of perovskite films, as well as the interfacial charge transport, strongly depends on the properties of hole transport layer materials at the buried interface.<sup>10,11</sup> Therefore, it is crucial to develop efficient charge transport layer materials, especially hole transport layer materials, to boost the photovoltaic performance of wide-bandgap PSCs.<sup>12–14</sup>

Self-assembled monolayer (SAM) materials featuring a carbazole core and phosphonic acid anchoring groups (e.g. 2-[(9H-carbazol-9-yl)ethyl]phosphonic acid (2PACz)<sup>15</sup> and [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz)<sup>16</sup>) have become a mainstream choice for constructing the buried interface in inverted PSCs, owing to their facile solution processability and excellent hole extraction capability. A series of

<sup>a</sup> State Key Laboratory of Powder Metallurgy, College of Chemistry and Chemical Engineering, Central South University, Changsha 410083, P.R. China.  
E-mail: zhaofuwen@csu.edu.cn

<sup>b</sup> CAS Key Laboratory of Nanosystem and Hierarchical Fabrication, CAS Center for Excellence in Nanoscience, National Center for Nanoscience and Technology, Beijing 100190, P. R. China. E-mail: liuxl2020@nanoctr.cn

<sup>c</sup> Beijing National Laboratory for Molecular Sciences, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, P. R. China

<sup>d</sup> College of Chemical Engineering, Beijing University of Chemical Technology, Beijing, 100029, China

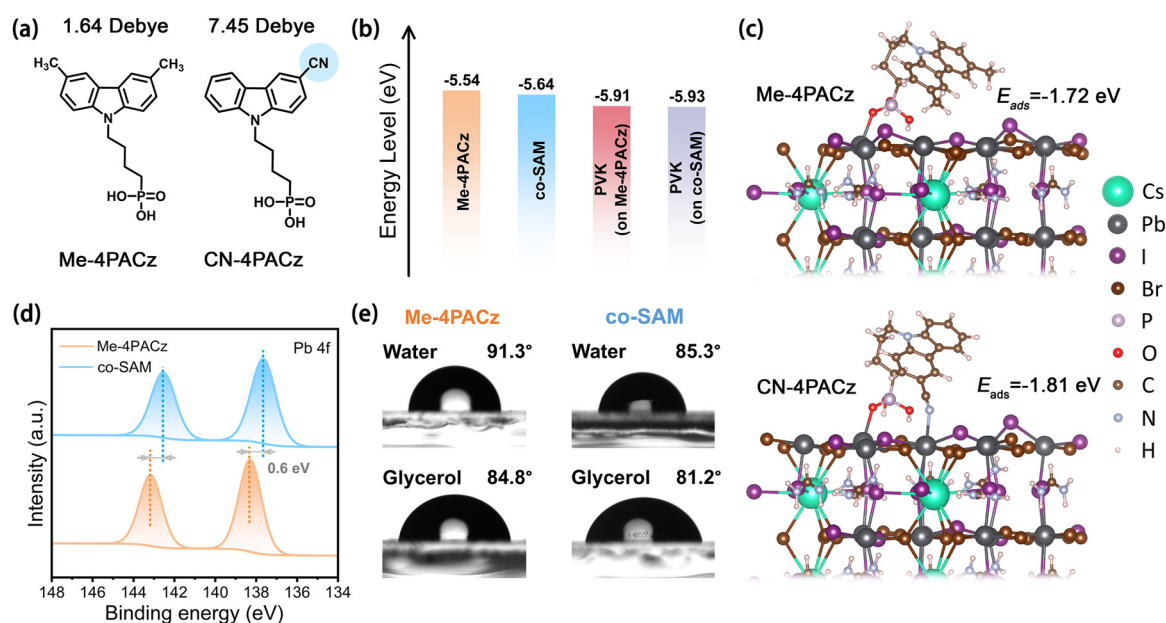
SAM molecules with diverse functional groups and properties have been developed,<sup>17,18</sup> successfully propelling the power conversion efficiency (PCE) of inverted PSCs to over 27% recently.<sup>19</sup> However, commonly used SAMs generally suffer from inhomogeneous distribution on electrodes<sup>20,21</sup> and inferior surface wettability for the perovskite precursor solution,<sup>22</sup> since amphiphilic SAMs tend to form micelles in polar solvents.<sup>23,24</sup> Strategies, such as shortening side chains,<sup>25</sup> introducing interfacial repairers,<sup>26</sup> adding co-solvents,<sup>21</sup> incorporating co-SAMs,<sup>27,28</sup> *etc.*, were employed to reduce the energetic barrier to disassemble the micelles and thus suppress self-aggregation for improved coverage. Recently, it has been reported that enlarging the dipole moment of SAMs can improve the molecular alignment, increase the work function of NiO<sub>x</sub>, and enhance the crystal quality of perovskite films, thus benefitting charge transport at the buried interface of PSCs.<sup>29–31</sup> Moreover, SAMs typically exhibit insufficient defect passivation capability and weak bonding strength to perovskite films, leading to notable interface leakage currents and non-radiative recombination.<sup>32,33</sup> Functional groups having strong interaction with perovskites, such as sulfhydryl,<sup>34</sup> methylthio,<sup>35</sup> methoxy,<sup>36</sup> *etc.*, were introduced into the SAM layer to strengthen the defect passivation at the anode interface. To date, nevertheless, research on developing SAMs that enhance wettability to improve perovskite film quality and strengthen defect passivation to suppress charge recombination for efficient wide-bandgap PSCs remains scarce.<sup>32,37</sup>

Recently, it has been reported that attaching cyano groups onto SAMs can enlarge the molecular dipole moment, increase hydrophilicity and improve the passivation ability to efficiently enhance the photovoltaic performance of PSCs.<sup>25,38–41</sup> Herein,

we developed a novel cyano-group modified SAM molecule, *viz.*, CN-4PACz (Fig. 1a) and integrated it with the classical Me-4PACz to form a co-self-assembled monolayer (co-SAM) for wide-bandgap PSCs. CN-4PACz has a deeper highest occupied molecular orbital (HOMO) energy level, larger molecular dipole moment and stronger interaction with the perovskite, due to the attachment of the strong electron-withdrawing cyano group. Incorporating CN-4PACz into Me-4PACz endows the co-SAM with improved energy level alignment with the perovskite, larger surface energy, enhanced defect passivation capacity, higher electrical conductivity and more uniform morphology, compared to those of the pristine Me-4PACz. Its impacts on the quality of perovskite films and charge carrier dynamics in wide-bandgap PSCs were investigated systematically. Benefitting from the superiority of the co-SAM, wide-bandgap PSCs having a co-SAM as the hole transport layer achieve a champion PCE of 18.41% with enhanced  $V_{OC}$  and fill factor (FF), which is among the highest values for PSCs with a wide-bandgap of > 1.86 eV. Moreover, unencapsulated devices retained 84.1% of their initial PCE after 1300 hours of storage under a nitrogen atmosphere, demonstrating their improved operational stability.

## Results and discussion

CN-4PACz was designed and incorporated into Me-4PACz to develop a co-assemble SAM by taking the following points into consideration: (1) the strong electron-withdrawing ability of the cyano group would endow CN-4PACz with a deeper HOMO energy level to improve the energy level alignment between the SAM and the perovskite film;<sup>42,43</sup> (2) the cyano group at the



**Fig. 1** (a) Chemical structures and dipole moments of Me-4PACz and CN-4PACz. (b) HOMO energy level diagrams of Me-4PACz, co-SAM, perovskite film on Me-4PACz and perovskite film on co-SAM. (c) Theoretical calculation for the absorption structure and adsorption energy of Me-4PACz and CN-4PACz on the perovskite surface. (d) XPS spectra for Pb 4f of the bottom surface of perovskite films deposited on Me-4PACz and co-SAM. (e) Contact angles of water (upper) and glycerol (bottom) on FTO/Me-4PACz and FTO/co-SAM films.

terminal has strong coordination capacity with  $\text{Pb}^{2+44-46}$  and can strengthen the interaction with the perovskite film, thus enhancing the defect passivation ability for the SAM; and (3) the larger molecular dipole moment of CN-4PACz would suppress the self-aggregation of the SAM to reduce voids at the buried interface.<sup>47,48</sup> Motivated by these, CN-4PACz was prepared, according to the synthetic route, as shown in Scheme S1. The synthetic details are provided in the SI. Nuclear magnetic resonance and mass spectra were employed to confirm the chemical structures of the intermediates and the target product, CN-4PACz (Fig. S1–S4).

The energy levels of Me-4PACz and CN-4PACz were initially estimated by using density functional theory (DFT) calculations. As shown in Fig. S5, the HOMO and the lowest unoccupied molecular orbital (LUMO) energy levels of CN-4PACz were calculated to be  $-5.85$  and  $-1.21$  eV, respectively, much deeper than those of Me-4PACz ( $-5.15/-0.62$  eV), as expected. Their HOMO energy levels were further confirmed by using cyclic voltammetry ( $-5.42$  eV for Me-4PACz *vs.*  $-5.74$  eV for CN-4PACz, Fig. S6) and ultraviolet photoelectron spectroscopy (UPS) ( $-5.54$  eV for Me-4PACz *vs.*  $-6.08$  eV for CN-4PACz, Fig. S7 and Table S1), which are highly consistent with the DFT results. Consequently, the HOMO energy level of co-SAM (Me-4PACz:CN-4PACz = 2:1, w/w) was determined to be reduced to  $-5.64$  eV (Fig. 1b and Fig. S7c). It should be noted that the shift in the energy level for SAMs does not translate into a significant change in the valence band of the perovskite layer (Fig. 1b). The deeper HOMO energy level of the co-SAM reduces the energy level mismatch between the hole transport layer and the perovskite film, which would be beneficial for hole extraction at the anode interface.

DFT calculations were also performed to assess the effect of incorporating CN-4PACz on the intermolecular interaction between SAMs and the perovskite film. As shown in Fig. 1c, the absorbed Me-4PACz and CN-4PACz lie on the perovskite film with a preferred configuration for defect passivation, as reported in the literature.<sup>22</sup> It reveals that Me-4PACz only has the P=O group to coordinate with  $\text{Pb}^{2+}$ , while CN-4PACz has both the P=O group and the cyano group to coordinate with  $\text{Pb}^{2+}$ . As a result, the adsorption energy ( $E_{\text{ads}}$ ) of CN-4PACz on the perovskite film is  $-1.81$  eV, higher than that of Me-4PACz on the perovskite film ( $-1.72$  eV), which signifies the stronger interaction between CN-4PACz and the perovskite film. X-ray photoelectron spectroscopy (XPS) measurement was further conducted at the exposed SAM interface after peeling off the perovskite layer to confirm the point. As shown in Fig. 1d and Fig. S8, the binding energies of Pb 4f and I 3d in perovskite films with co-SAMs shift to a lower energy by 0.6 and 0.2 eV, respectively, compared to the Me-4PACz samples. The pronounced shifts for Pb 4f and I 3d verify the enhanced interaction between the co-SAM and perovskite films.<sup>49</sup> Furthermore, it can be expected that incorporating CN-4PACz into Me-4PACz would afford an improved passivation capacity for co-SAMs. Fourier transform infrared spectroscopy of Me-4PACz, co-SAM, Me-4PACz +  $\text{PbI}_2$  and co-SAM +  $\text{PbI}_2$  was also performed to evaluate the interaction between SAMs and the

perovskite (Fig. S9). It can be found that the stretching vibration of P=O slightly shifts from  $1275.23\text{ cm}^{-1}$  (Me-4PACz) to  $1274.31\text{ cm}^{-1}$  (Me-4PACz +  $\text{PbI}_2$ ), while the stretching vibration of P=O slightly shifts from  $1275.30\text{ cm}^{-1}$  (co-SAM) to  $1273.25\text{ cm}^{-1}$  (co-SAM +  $\text{PbI}_2$ ) and the stretching vibration of C≡N slightly shifts from  $2223.77\text{ cm}^{-1}$  (co-SAM) to  $2220.72\text{ cm}^{-1}$  (co-SAM +  $\text{PbI}_2$ ). This signifies that Me-4PACz and CN-4PACz can co-absorb onto the perovskite, which is consistent with the XPS results provided above.

According to the DFT result, the molecular dipole moment of CN-4PACz is 7.45 D, much larger than that of Me-4PACz (1.64 D), as expected (Fig. 1a). It originates from the strong electron-withdrawing capacity of cyano groups and the asymmetry of molecular skeleton. The incorporation of CN-4PACz with a large molecular dipole moment would affect the surface energy and molecular stacking.<sup>50,51</sup> The surface energy of SAM on fluorine-doped tin oxide (FTO) was investigated by using the contact angle method. As shown in Fig. 1e, contact angles of water and glycerol on FTO/Me-4PACz and FTO/co-SAM were  $91.3^\circ/84.8^\circ$  and  $85.3^\circ/81.2^\circ$ , respectively. As presented in Table S2, the calculated surface energy for co-SAM is higher than that for Me-4PACz ( $27.16$  *vs.*  $24.20\text{ mN m}^{-1}$ ), which can be ascribed to the enhanced polar component ( $17.14$  *vs.*  $13.15\text{ mN m}^{-1}$ ). In general, the higher surface energy would facilitate the SAM to adopt a two-dimensional layer-by-layer growth mode, which not only benefits the systematic arrangement of SAMs, but also fosters favorable charge transport dynamics within the perovskite device.<sup>52</sup> Thus, introducing CN-4PACz would endow the co-SAM with favorable molecular stacking and thus facilitate charge carrier transport at interfaces.

To verify the points above, we further investigated the morphology and electrical properties of Me-4PACz and co-SAM thin films on FTO *via* atomic force microscopy (AFM) and Kelvin probe force microscopy (KPFM), respectively. As shown in Fig. 2a–c, AFM height images present that the root-mean-square (RMS) roughness on FTO decreases from 21.8 nm (Me-4PACz) to 19.9 nm (co-SAM) and the distribution of surface height variations becomes more concentrated after incorporating CN-4PACz. The corresponding KPFM surface potential mapping in Fig. 2d and e demonstrates that the work function of the co-SAM (5.35 eV) is slightly larger than that of the Me-4PACz film (5.33 eV), due to the deeper HOMO of CN-4PACz for the strong electron-withdrawing cyano group. More importantly, the spatial uniformity of the surface potential of co-SAM was also significantly improved, compared to that of the Me-4PACz film. The AFM and KPFM results confirm a more uniform morphology for the co-SAM, which reflects that CN-4PACz can mitigate the molecular aggregation in the co-SAM. The electrical conductivity of Me-4PACz and co-SAM was further assessed by fabricating devices with the structure of FTO/Me-4PACz or co-SAM/Ag. As shown in Fig. S10, the co-SAM-based device exhibits a higher electrical conductivity of  $1.59 \times 10^{-5}\text{ S cm}^{-1}$  compared to the Me-4PACz based devices ( $7.68 \times 10^{-6}\text{ S cm}^{-1}$ ). The enhanced electrical conductivity of the co-SAM would facilitate charge carrier transport and collection.

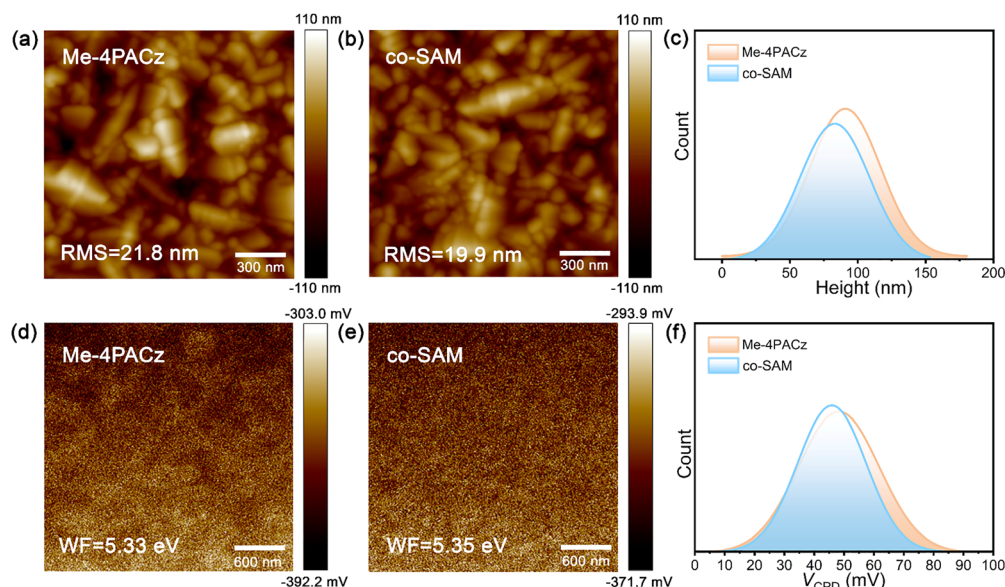


Fig. 2 AFM height images of (a) Me-4PACz and (b) co-SAM. (c) Statistical distribution comparison of surface heights for the two SAM layers. KPFM images of (d) Me-4PACz and (e) co-SAM. (f) Statistical distribution comparison of contact potential difference (CPD) for the two SAM layers.

The favorable co-SAM would lay a solid foundation for the uniform nucleation and high-quality growth of subsequent perovskite layers.<sup>53</sup> To compare the crystalline quality of perovskite films on Me-4PACz and co-SAM, we employed scanning electron microscopy (SEM) to investigate the bottom and top surface morphologies of perovskite films. As shown in Fig. 3a, distinct cracks and pinholes are observed at the bottom of the perovskite film deposited on Me-4PACz. These cracks and pinholes usually act as charge carrier recombination centers, leading to inferior device efficiency and stability.<sup>54</sup> In contrast, the bottom of the perovskite film deposited on the co-SAM exhibits a continuously dense and pinhole-free morphology,

which should be attributed to the enhanced coordination of CN-4PACz and the homogeneous packing of the co-SAM. Furthermore, top-surface images revealed that the perovskite/co-SAM sample has a larger grain size (432 nm) and a reduced root-mean-square (RMS) roughness (12.9 nm), compared with the control perovskite film (343 nm for grain size and 13.6 nm for RMS), as shown in Fig. 3c–3f and S11. This indicates that the integration of CN-4PACz with Me-4PACz as the SAM can improve the crystalline quality of the perovskite film, which would help reduce the non-radiative recombination for perovskite PSCs. Then, steady-state photoluminescence (PL) spectroscopy was also measured to study the carrier dynamics for the

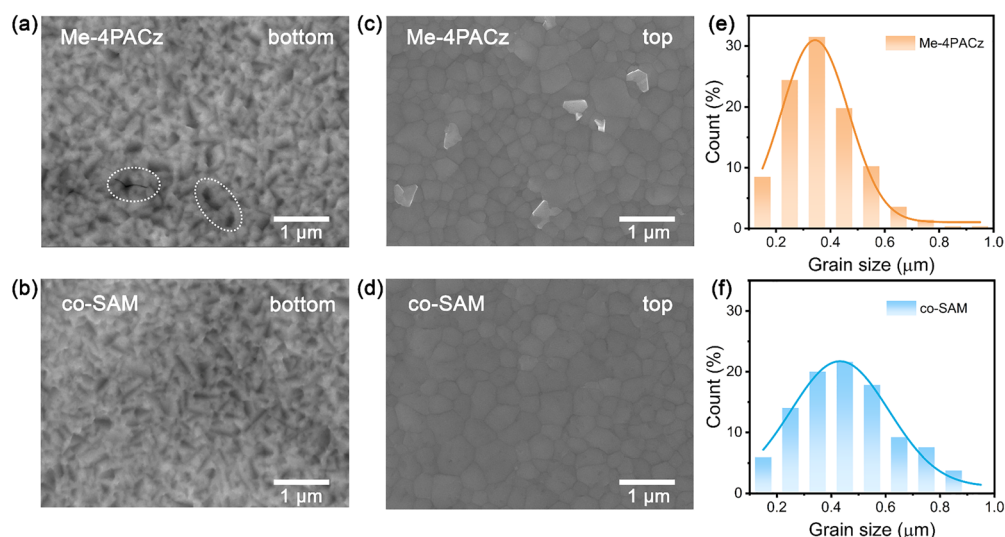
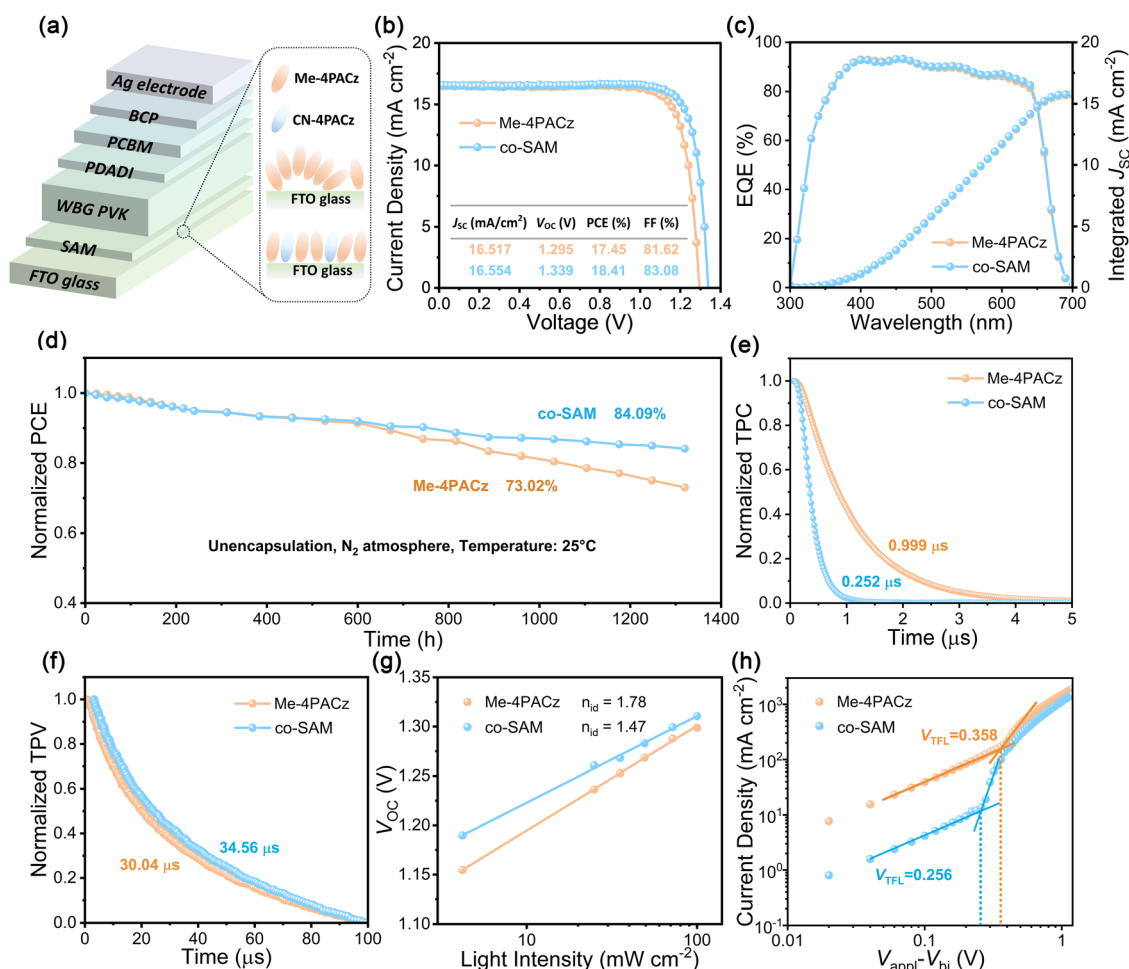


Fig. 3 SEM images of the buried interface of perovskite films deposited on (a) Me-4PACz and (b) co-SAM. SEM images of the top surface of perovskite films deposited on (c) Me-4PACz and (d) co-SAM. Histograms showing the particle size distribution at the top surface of perovskite thin films deposited on (e) Me-4PACz and (f) co-SAM.

perovskite film. As shown in Fig. S12, the PL intensity of the perovskite/co-SAM sample was obviously enhanced compared to that of perovskite/Me-4PACz, indicating the suppressed non-radiative recombination for the enhanced quality of the perovskite film.<sup>55</sup>

According to the results mentioned above, the co-SAM shows great potential for wide-bandgap PSCs. Motivated by this, we fabricated typical inverted wide-bandgap PSCs with the configuration of FTO/SAM/FA<sub>0.75</sub>CS<sub>0.25</sub>Pb(I<sub>0.5</sub>Br<sub>0.5</sub>)<sub>3</sub>/PDADI/PCBM/BCP/Ag (noted that PDADI is 1,3-diaminopropane dihydroiodide and BCP is bathocuproine) (Fig. 4a). The photocurrent density–voltage (*J*–*V*) curves and the corresponding photovoltaic parameters of best-performing PSCs are presented in Fig. 4b. Compared with the control PSC under the reverse scan, the target device exhibits an increase in PCE from 17.45% to 18.41%, mainly due to the enhanced *V*<sub>OC</sub> (1.295 vs. 1.339 V) and FF (81.62% vs. 83.08%). Note that the PCE of 18.41% is among the highest values for wide-bandgap (> 1.86 eV) PSCs so far. Consequently, the *V*<sub>OC</sub> loss decreases from 0.571 to 0.526 V,

according to bandgaps calculated from the EQE spectra (Fig. S13). In addition, the hysteresis factor decreases from 5.3% to 3.5%, attributable to the enhanced crystalline quality of the perovskite film based on the co-SAM (Fig. S14 and Table S3). We also tracked the control and target devices at their maximum power points for 550 s to obtain the steady-state output power curves (Fig. S15). The stabilized efficiency of the target device is 18.38%, while that of the control device is 17.32%, both of which are similar to the PCEs measured above. The external quantum efficiency (EQE) curves in Fig. 4c show that the integrated current densities of target and control devices are 15.82 and 15.73 mA cm<sup>−2</sup>, respectively, both consistent with the *J*<sub>SC</sub>s from the *J*–*V* results (mismatch < 5%). The narrower statistical distributions of photovoltaic parameters for the target devices, in Fig. S16 and Table S4, demonstrate that the co-SAM can improve the reproducibility of PSCs. Additionally, the long-term operational reliability of the devices was also evaluated. As shown in Fig. 4d, the device based on the co-SAM retains 84.09% of its initial PCE, while the control device



**Fig. 4** (a) Schematic diagram of device configuration of inverted wide-bandgap PSCs. (b) Champion *J*–*V* curves and photovoltaic parameters of wide-bandgap PSCs (*E*<sub>g</sub> = 1.87 eV) based on Me-4PACz and co-SAM. (c) EQE spectra and the integrated *J*<sub>SC</sub> of the best-performing devices based on Me-4PACz and co-SAM with an active area of 0.09 cm<sup>2</sup>. (d) The normalized PCE evolution of unencapsulated PSCs based on Me-4PACz and co-SAM stored in a nitrogen atmosphere. (e) TPC and (f) TPV curves of PSCs based on Me-4PACz and co-SAM. (g) The dependence of *V*<sub>OC</sub> on light intensity of the target and control PSCs. (h) SCLC of the hole-only devices based on Me-4PACz and co-SAM.

maintains only 73.02% after storage in a N<sub>2</sub> environment for over 1300 hours. The improvement in stability can be attributed to the uniform morphology of the co-SAM layer, which effectively suppresses interfacial ion migration and degradation, thereby enhancing the intrinsic robustness of the device.

To elucidate the underlying mechanisms for the improved  $V_{OC}$  and FF of the target device, charge carrier dynamics were investigated using transient photocurrent (TPC) and transient photovoltage (TPV). As shown in Fig. 4e, the device based on the co-SAM exhibits a shorter charge extraction time ( $\tau_C = 0.252 \mu s$ ) compared to the Me-4PACz-based device ( $\tau_C = 0.999 \mu s$ ), which indicates a quicker charge carrier transport and extraction in the target device. The TPV results in Fig. 4f show that the charge carrier lifetime increases from 30.04 to 34.56  $\mu s$  after integrating CN-4PACz with Me-4PACz. The ideal factor  $n_{id}$  obtained *via* fitting the curves of Fig. 4g represents the trap-assisted charge recombination in PSCs. The reduced  $n_{id}$  (1.47) for the target PSC confirms that the trap-assisted charge recombination is significantly suppressed by using co-SAMs. Furthermore, dark  $J-V$  measurements reveal a substantial reduction in the leakage current for the target device, as illustrated in Fig. S17. In electrochemical impedance spectroscopy (EIS) analysis provided in Fig. S18 and Table S5, the co-SAM-based device exhibits a smaller series resistance ( $R_s$ ) than the Me-4PACz-based one (20.38 *vs.* 38.96  $\Omega \text{ cm}^2$ ). The reduction can be ascribed to the improved ohmic contact and more efficient charge transport for the uniform and stronger interaction with the perovskite for the co-SAM layer. Concurrently, the charge transfer resistance ( $R_{ct}$ ) increased substantially from 713 to 1745  $\Omega \text{ cm}^2$ . The increase in  $R_{ct}$  confirms that the non-radiative recombination process has been effectively suppressed in the perovskite bulk and at the interface of the co-SAM-based device.<sup>56</sup>

To quantitatively analyze the trap density in the perovskite devices, space-charge-limited current (SCLC) measurements were conducted on the hole-only devices with a structure of FTO/Me-4PACz or co-SAM/perovskite/spiro-OMeTAD/Ag, as shown in Fig. 4h. The target device based on the co-SAM exhibits a trap-filled limit voltage ( $V_{TFL}$ ) of 0.256 V with a trap state density of  $2.12 \times 10^{15} \text{ cm}^{-3}$ , which is lower than the  $V_{TFL}$  of 0.358 V with a trap state density of  $2.97 \times 10^{15} \text{ cm}^{-3}$  for the control device. The reduced trap density is attributed to the strengthened defect passivation capacity of the co-SAM and improved morphology of the co-SAM-based device. Therefore, the trap-assisted charge recombination is suppressed, contributing to the higher  $V_{OC}$  and FF for enhanced PCEs of PSCs with the co-SAM.

## Conclusion

In summary, we synthesized a cyanation-functionalized SAM molecule, CN-4PACz, and integrated it with the classical SAM molecule, Me-4PACz, to develop a co-SAM for wide-bandgap PSCs. The cyano group on CN-4PACz endows it with a deeper HOMO energy level (−6.08 eV), larger molecular dipole

moment (7.45 D) and stronger interaction with the perovskite. Incorporating CN-4PACz into Me-4PACz enables the co-SAM to acquire improved energy level alignment with the perovskite, larger surface energy, enhanced defect passivation capacity, higher electrical conductivity and more uniform morphology compared to those of the pristine Me-4PACz. Thus, perovskite films deposited on the co-SAM layer present improved crystal quality with a larger grain size, a smoother surface and reduced interfacial defects, which help suppress trap-assisted recombination and benefit charge transport and extraction in devices. Consequently, wide-bandgap ( $\sim 1.87 \text{ eV}$ ) PSCs having the co-SAM layer achieve a champion PCE of 18.41% with a higher  $V_{OC}$  of 1.339 V and an FF of 83.08% compared to the ones having Me-4PACz (17.45%). The PCE is among the highest values for wide-bandgap ( $> 1.86 \text{ eV}$ ) PSCs. Besides, the operational stability is also enhanced for the co-SAM-based PSCs. This work highlights an effective interfacial engineering approach to boost the photovoltaic performance of wide-bandgap PSCs.

## Conflicts of interest

The authors declare no conflicts of interest.

## Data availability

The data supporting this article have been included as part of the supplementary information (SI). Supplementary information: Tables S1–S5, NMR spectra, DFT calculation, cyclic voltammetry, ultraviolet photoelectron spectroscopy (UPS), electrical conductivity, contact angle, AFM images, photoluminescence spectra, EQE measurement, photovoltaic parameters, dark  $J-V$  curve, electrochemical impedance spectroscopy and further experimental details. See DOI: <https://doi.org/10.1039/d6tc00781c>.

## Acknowledgements

This work was supported by the National Natural Science Foundation of China (Grant No. 52372056, 52232003, 22422901, and 22379010).

## References

- 1 J. Han, K. Park, S. Tan, Y. Vaynzof, J. Xue, E. W. G. Diau, M. G. Bawendi, J. W. Lee and I. Jeon, *Nat. Rev. Methods Primers*, 2025, 5, 3.
- 2 A. K. Astakala, S. Y. Lee, J. Gautam, K. B. Thapa, I. In, S. J. Lee and S. J. Park, *Adv. Powder Mater.*, 2026, 5, 100354.
- 3 X. Wu, D. Zhang, B. Liu, Y. Wang, X. Wang, Q. Liu, D. Gao, N. Wang, B. Li, L. Wang, Z. Yu, X. Li, S. Xiao, N. Li, M. Stollerfoht, Y.-H. Lin, S. Yang, X. C. Zeng and Z. Zhu, *Adv. Mater.*, 2024, 36, 2410692.
- 4 Y. Han, J. Fu, Z. Ren, J. Yu, Q. Liang, Z. Xu, X. Xie, D. Li, R. Ma, M. Cao, Y. Sun, C. Yang, J. He, X. Chang, K. Liu, P. W. K. Fong, J. Huang, H. Liu, Z. Liu, D. Xu, L. Cheng,

- J. Zhang, G. Yang, X. Lu, Y. Zhu, Q. Tai, Q. Lin, H. Hu, Y. Yang and G. Li, *Nat. Energy*, 2025, **10**, 513–525.
- 5 Y. Li, Y. Yan, Y. Fu, W. Jiang, M. Liu, M. Chen, X. Huang, G. Lu, X. Lu, J. Yin, S. Wu and A. K. Y. Jen, *Angew. Chem., Int. Ed.*, 2024, **63**, e202412515.
- 6 K. O. Brinkmann, P. Wang, F. Lang, W. Li, X. Guo, F. Zimmermann, S. Olthof, D. Neher, Y. Hou, M. Stolterfoht, T. Wang, A. B. Djurišić and T. Riedl, *Nat. Rev. Mater.*, 2024, **9**, 202–217.
- 7 N. Yan, Y. Gao, J. Yang, Z. Fang, J. Feng, X. Wu, T. Chen and S. Liu, *Angew. Chem., Int. Ed.*, 2023, **62**, e202216668.
- 8 E. Aydin, T. G. Allen, M. De Bastiani, A. Razzaq, L. Xu, E. Ugur, J. Liu and S. D. Wolf, *Science*, 2024, **383**, eadh3849.
- 9 F. Xu, M. Zhang, Z. Li, X. Yang and R. Zhu, *Adv. Energy Mater.*, 2023, **13**, 2203911.
- 10 Q. Jiang, Y. Zhao, X. Zhang, X. Yang, Y. Chen, Z. Chu, Q. Ye, X. Li, Z. Yin and J. You, *Nat. Photonics*, 2019, **13**, 460–466.
- 11 H. Cui, L. Huang, S. Zhou, C. Wang, X. Hu, H. Guan, S. Wang, W. Shao, D. Pu, K. Dong, J. Zhou, P. Jia, W. Wang, C. Tao, W. Ke and G. Fang, *Energy Environ. Sci.*, 2023, **16**, 5992–6002.
- 12 T. Liu, C. Luo, R. He, Z. Zhang, X. Lin, Y. Chen and T. Wu, *Adv. Mater.*, 2025, **37**, 2502032.
- 13 L. Pan, R. Yu, Z. Fang, B. Chen, L. Zheng, R. Hu and G. Chen, *Chem. Eng. J.*, 2025, **525**, 170528.
- 14 Y. Zhang, Y. Liu, Z. Zhao, T. Kong, W. Chen, W. Liu, P. Gao and D. Bi, *Adv. Funct. Mater.*, 2025, **35**, 2417575.
- 15 A. Al-Ashouri, A. Magomedov, M. Roß, M. Jošt, M. Talaikis, G. Chistiakova, T. Bertram, J. A. Márquez, E. Köhnen, E. Kasparavičius, S. Levchenko, L. Gil-Escrig, C. J. Hages, R. Schlattmann, B. Rech, T. Malinauskas, T. Unold, C. A. Kaufmann, L. Korte, G. Niaura, V. Getautis and S. Albrecht, *Energy Environ. Sci.*, 2019, **12**, 3356–3369.
- 16 A. Al-Ashouri, E. Köhnen, B. Li, A. Magomedov, H. Hempel, P. Caprioglio, J. A. Márquez, A. B. M. Vilches, E. Kasparavičius, J. A. Smith, N. Phung, D. Menzel, M. Grischek, L. Kegelmann, D. Skroblin, C. Gollwitzer, T. Malinauskas, M. Jošt, G. Matič, B. Rech, R. Schlattmann, M. Topić, L. Korte, A. Abate, B. Stannowski, D. Neher, M. Stolterfoht, T. Unold, V. Getautis and S. Albrecht, *Science*, 2020, **370**, 1300–1309.
- 17 M. Azam, T. Du, Z. Wan, H. Zhao, H. Zeng, R. Wei, C. J. Brabec, J. Luo and C. Jia, *Energy Environ. Sci.*, 2024, **17**, 6974–7016.
- 18 Q. Jiang and K. Zhu, *Nat. Rev. Mater.*, 2024, **9**, 399–419.
- 19 Z. Xiong, Q. Zhang, K. Cai, H. Zhou, Q. Song, Z. Han, S. Kang, Y. Li, Q. Jiang, X. Zhang and J. You, *Science*, 2025, **390**, 638–642.
- 20 Z. Yi, W. Wang, R. He, J. Zhu, W. Jiao, Y. Luo, Y. Xu, Y. Wang, Z. Zeng, K. Wei, J. Zhang, S. W. Tsang, C. Chen, W. Tang and D. Zhao, *Energy Environ. Sci.*, 2024, **17**, 202–209.
- 21 M. Liu, L. Bi, W. Jiang, Z. Zeng, S. W. Tsang, F. R. Lin and A. K. Y. Jen, *Adv. Mater.*, 2023, **35**, 2304415.
- 22 S. Liu, J. Li, W. Xiao, R. Chen, Z. Sun, Y. Zhang, X. Lei, S. Hu, M. K. Czerny, J. Wang, F. Ren, Q. Zhou, H. Raza, Y. Gao, Y. Ji, S. Li, H. Li, L. Qiu, W. Huang, Y. Zhao, B. Xu, Z. Liu, H. J. Snaith, N. G. Park and W. Chen, *Nature*, 2024, **632**, 536–542.
- 23 A. Ulman, *Chem. Rev.*, 1996, **96**, 1533–1554.
- 24 A. Ullah, K. H. Park, H. D. Nguyen, Y. Siddique, S. Shah, H. Tran, S. Park, S. I. Lee, K. K. Lee and C. H. Han, *Adv. Energy Mater.*, 2022, **12**, 2103175.
- 25 S. Zhang, F. Ye, X. Wang, R. Chen, H. Zhang, L. Zhan, X. Jiang, Y. Li, X. Ji, S. Liu, M. Yu, F. Yu, Y. Zhang, R. Wu, Z. Liu, Z. Ning, D. Neher, L. Han, Y. Lin, H. Tian, W. Chen, M. Stolterfoht, L. Zhang, W.-H. Zhu and Y. Wu, *Science*, 2023, **380**, 404–409.
- 26 C.-M. Hung, C.-C. Wu, Y.-H. Yang, B.-H. Chen, C.-H. Lu, C.-C. Chu, C.-H. Cheng, C.-Y. Yang, Y.-D. Lin, C.-H. Cheng, J.-Y. Chen, I.-C. Ni, C.-I. Wu, S.-D. Yang, H.-C. Chen and P.-T. Chou, *Adv. Sci.*, 2024, **11**, 2404725.
- 27 Z. Min, C. Guo, J. Guo, Y. Kong, X. Hao, Z. Tang, Y. Li, W. Ma and J. Yuan, *Adv. Funct. Mater.*, 2025, DOI: [10.1002/adfm.202528728](https://doi.org/10.1002/adfm.202528728).
- 28 X. Zhang, Y. Wang, K. Zhang, M. Tao, H. Guo, L. Guo, Z. Song, J. Wen, Y. Yang, Y. Hou and Y. Song, *Angew. Chem., Int. Ed.*, 2025, **64**, e202423827.
- 29 S. Wang, D. Khan, W. Zhou, Y. Sui, T. Zhang, G. Yu, Y. Huang, X. Yang, X. Chen, H. Yan, J. Tang, F. Yang, P. Han, Z. Zheng, Y. Zhang and Z. Tang, *Adv. Funct. Mater.*, 2024, **34**, 2316202.
- 30 H. Gao, D. Khan, I. Muhammad, Y. Zhang, C. Zhang, P. Tong, J. Wang, X. Lai, D. Lu, I. Ghani, W. Chen, J. Tang, P. You, X. Yang and Z. Tang, *Adv. Funct. Mater.*, 2026, **36**, e19073.
- 31 I. Ghani, S. Tingshu, S. Ahmed, Z. Hongbin, Z. Zirun, Y. Zhang, P. You, I. Muhammad, T. Zeguo and D. Khan, *Adv. Sci.*, 2026, **13**, e18676.
- 32 S. Fu, N. Sun, S. Hu, H. Chen, X. Jiang, Y. Li, X. Zhu, X. Guo, W. Zhang, X. Li, A. S. Vasenko and J. Fang, *Energy Environ. Sci.*, 2025, **18**, 5503–5510.
- 33 S. M. Park, M. Wei, N. Lempesis, W. Yu, T. Hossain, L. Agosta, V. Carnevali, H. R. Atapattu, P. Serles, F. T. Eickemeyer, H. Shin, M. Vafaie, D. Choi, K. Darabi, E. D. Jung, Y. Yang, D. B. Kim, S. M. Zakeeruddin, B. Chen, A. Amassian, T. Filleter, M. G. Kanatzidis, K. R. Graham, L. Xiao, U. Rothlisberger, M. Grätzel and E. H. Sargent, *Nature*, 2023, **624**, 289–294.
- 34 C. Peng, X. Sun, B. Zhang, H. Meng, K. Wang, Z. Li, M. Zhu, X. Zhang, G. Wang, Z. Zhou and S. Pang, *J. Am. Chem. Soc.*, 2025, **147**, 48391–48399.
- 35 C.-H. Chen, G.-W. Liu, X. Chen, C. Deger, R.-J. Jin, K.-L. Wang, J. Chen, Y. Xia, L. Huang, I. Yavuz, J. Fan and Z.-K. Wang, *Angew. Chem., Int. Ed.*, 2025, **64**, e202419375.
- 36 D. Zhang, B. Yan, R. Xia, B. Li, R. Li, P. Hang, H. Xin, J. Wei, M. Lei, Y. Chen, J. Gao, H. Zhang, Z. Ni, D. Yang and X. Yu, *Nat. Photonics*, 2026, **20**, 40–48.
- 37 J. G. Son, S. Ameen, J. Roe, S. Park, J. Seo, J. Kim, A. B. Faheem, H. Koo, S. O. Oh, Y. Jo, J. W. Kim, Y. J. Lee, Y. S. Shin, H. Jang, D. Lee, S. H. Hur, K.-K. Lee, S. Cho, D. S. Kim, J. Y. Kim and B. S. Kim, *Adv. Energy Mater.*, 2025, **15**, 2404092.

- 38 X. Zhou, Y. Sun, B. Tang, F. Wang, X. Ding, H. Lin, Y. Tong, X. Liu, H. Sun, Q. Zhu, C. Yang, Z. Ren, M. Yuan, G. Li, H. Zhang and H. Hu, *Angew. Chem., Int. Ed.*, 2025, **64**, e18169.
- 39 P. Ferdowsi, E. Lee, G. Jang, J. S. Park, D. Lee, S. K. Sharma, W. Thor, J.-W. Ha, H.-H. Cho, J.-H. Yum and K. Sivula, *Adv. Energy Mater.*, 2025, **15**, e02789.
- 40 Y. Zhong, Z. Fang, N. Ma, S. Zhang, L. Shen, C. Sun, H. Feng, Y. Jin, Y. Li, L. Xie, Y. Wu and Z. Wei, *Adv. Funct. Mater.*, 2025, DOI: [10.1002/adfm.202525359](https://doi.org/10.1002/adfm.202525359).
- 41 Q. Liao, Y. Wang, Z. Zhang, K. Yang, Y. Shi, K. Feng, B. Li, J. Huang, P. Gao and X. Guo, *J. Energy Chem.*, 2022, **68**, 87–95.
- 42 M. Y. Kuo, H. Y. Chen and I. Chao, *Chemistry*, 2007, **13**, 4750–4758.
- 43 Y. Li, E. Huang, X. Guo and K. Feng, *Mater. Chem. Front.*, 2023, **7**, 3803–3819.
- 44 N. K. Noel, A. Abate, S. D. Stranks, E. S. Parrott, V. M. Burlakov, A. Goriely and H. J. Snaith, *ACS Nano*, 2014, **8**, 9815–9821.
- 45 S. Fu, N. Sun, H. Chen, C. Liu, X. Wang, Y. Li, A. Abudulimu, Y. Xu, S. Ramakrishnan, C. Li, Y. Yang, H. Wan, Z. Huang, Y. Xian, Y. Yin, T. Zhu, H. Chen, A. Rahimi, M. M. Saeed, Y. Zhang, Q. Yu, D. S. Ginger, R. J. Ellingson, B. Chen, Z. Song, M. G. Kanatzidis, E. H. Sargent and Y. Yan, *Nat. Nanotechnol.*, 2025, **20**, 772–778.
- 46 Y. Lv, C. Yang, Z. Su, T. He, R. Wang, R. Chen, X. Gao and W.-H. Zhang, *Energy Environ. Sci.*, 2025, **18**, 6237–6247.
- 47 Z. Chen, X. Li, S. Qin, Y. Gong, Z. Liu, M. Yuan, T. Liang, L. Meng and Y. Li, *CCS Chem.*, 2024, **7**, 507–518.
- 48 W. Jiang, F. Li, M. Li, F. Qi, F. R. Lin and A. K. Y. Jen, *Angew. Chem., Int. Ed.*, 2022, **61**, e202213560.
- 49 C. Li, Y. Chen, Y. Li, L. Gong, Z. Yuan, L. Liang, J. Chen, P. Ganesan, Y. Zhang, J. Ma and P. Gao, *Angew. Chem., Int. Ed.*, 2025, **64**, e202420585.
- 50 Y. Shi, Y. Zhang, H. Guo, Y. Zhang, M. Li, C. Xiao, W. Huang, Y. Zhai, J. Xia, E. Aydin and F. Zhang, *Joule*, 2025, **9**, 102009.
- 51 C. Li and P. Gao, *Chin. J. Struct. Chem.*, 2024, **43**, 100324.
- 52 Z. Yi, X. Li, Y. Xiong, G. Shen, W. Zhang, Y. Huang, Q. Jiang, X. R. Ng, Y. Luo, J. Zheng, W. L. Leong, F. Fu, T. Bu and J. Yang, *Interdiscip. Mater.*, 2024, **3**, 203–244.
- 53 Y. Peng, Y. Chen, J. Zhou, C. Luo, W. Tang, Y. Duan, Y. Wu and Q. Peng, *Nat. Commun.*, 2025, **16**, 1252.
- 54 Z. Liang, Y. Zhang, H. Xu, W. Chen, B. Liu, J. Zhang, H. Zhang, Z. Wang, D.-H. Kang, J. Zeng, X. Gao, Q. Wang, H. Hu, H. Zhou, X. Cai, X. Tian, P. Reiss, B. Xu, T. Kirchartz, Z. Xiao, S. Dai, N.-G. Park, J. Ye and X. Pan, *Nature*, 2023, **624**, 557–563.
- 55 W. Jiao, Y. Song, J. Wang, J. You, Y. Xu, D. Ma, H. Huang, J. Zeng, Y. Luo, J. Zhu, Z. Zhang, Z. Yi, S. Wu, Z. Gao, J. Jin, W. Wang, X. Hao, G. Zeng, B. Zou, Y. Zhang, P. Yang, B. Xu, C. Chen, S. Ren and D. Zhao, *Adv. Energy Mater.*, 2025, **15**, 2501556.
- 56 J. Guo, G. Meng, X. Zhang, H. Huang, J. Shi, B. Wang, X. Hu, J. Yuan and W. Ma, *Adv. Mater.*, 2023, **35**, 2302839.